

Experiment 'p_1_2_sum' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_2_sum

Natural language statement: Write a method that returns the sum of the first ten positive integers, $1 + 2 + \dots + 10$.

Method signature: p_1_2_sum() returns (i: int)

Ensures

- $i == (10 * 11) / 2$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that returns the sum of the first ten positive integers, $1 + 2 + \dots + 10$.

The signature should be:

`p_1_2_sum()` returns `(i: int)`

The method should respect the following contract:

`ensures i == (10 * 11) / 2`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdL7pkik34BX8oatarD62p5cbxQ2

Model: gpt-5.5-2026-04-23

Created at: 1780483337

Finish reason: stop

Usage: 280 tokens in, and 85 tokens out

```
{  
  i := (10 * 11) / 2;  
}
```

Final Program

```
method p_1_2_sum() returns (i: int)  
  ensures i == (10 * 11) / 2  
{  
  i := (10 * 11) / 2;  
}
```

Total Token Usage

Input tokens: 280

Output tokens: 85

Reasoning tokens: 51

Sum of ‘total tokens’: 365

Experiment Timings

Overall Experiment started at 1780483337431, ended at 1780483344423, lasting 6992ms (6.99 seconds)

Iteration #1 started at 1780483337432, ended at 1780483344423, lasting 6991ms (6.99 seconds)